Data Analytics Report

Week 3 – Trained Models and Prediction Report

AI Powered Data Analysis Remote Internship

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Team name: Team 6

Dataset: Student Participation Dataset

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Contributions

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Feature Importance Analysis – Busayo Saulu & Genevieve Ah Shene

Opportunity Category Performance – Chiamaka Chinaza

Quarter Demand Forecast – Chiamaka Chinaza & Usman Bin Yosuf

Model Evaluation - Chiamaka Chinaza & Genevieve Ah Shene

Forecast Visualization – Yonnie Ntsubane

XGBoost Model build – Sudheer Muthyala

Introduction

For this deliverable, our focus was on predictive machine learning, building on the EDA, visualization, and other forecasting insights we had already discovered. We leveraged relevant historical data on learner signups, participation, and engagement to make sense of patterns and predict potential participation trends and behaviors relevant to the different opportunity categories.

We evaluated the trends in quarterly opportunities and participation, identifying peak and low seasons for different categories of opportunities. Using machine learning algorithms, we also created probability scores and identified associated factors that might affect a learner's decision to participate in an opportunity, including progression, engagement, category, and demographics. Feature engineering was also done to enable the algorithms to recognize patterns linked to years, months, and quarters. Our final outputs were visualized to highlight key insights, trends, and predictions.

We developed two different machine learning algorithms (Random Forest and XGBoost) to evaluate which produced better predications for our different analysis tasks. Specifically, Random Forest was our baseline ensemble learning model, and XGBoost was our boosting-based algorithm comparison model. We compared performance based on accuracy, precision, recall, F1-score, AUC predicted volumes and feature importance depending on the predication task.

Overall, our analysis provided a strong foundation for further investigations on the topic or informing stakeholders of potential peak seasons, low seasons, and influential factors that play a role in learner participation trends and behaviors.

Feature Importance Analysis

1. Most Influential variables

Using 5 – fold cross validation, the performance of Random Forest and XGBoost models was compared across different feature sets.

Feature Set	Number	RF AUC	RF Accuracy	XGB AUC	XGB Accuracy
All features	27	0.9930	96.81%	0.9950	98.93%
Top 8	8	0.9928	97.3%	0.9942	97.6%
Top 5	5	0.9929	97.3%	0.9943	97.5%
Top 3	3	0.9882	94.9%	0.9854	93.3%
Learner only	8	0.7041	65.5%	0.7874	71.5%

Comparison: XGBoost achieved the highest performance when all features were included, with an AUC of 0.9944 and accuracy of 97.5%, compared with 0.9930 and 96.8% for Random Forest. However, after reducing the dataset to the top five features, both models achieved the same accuracy of 97.3%. This suggests that a smaller group of influential variables can retain most of the predictive performance.

The 5 -fold cross validation ensured that the data was split into 5 parts: model was trained on four and tested on one.

The eight features are:

1. Days between opportunity start and apply date
2. Program duration (months)
3. Program duration (days)

4. Opportunity Category
5. Months between start and apply date
6. Opportunity end month
7. Opportunity starts year
8. Opportunity starts month

The five features are:

1. Days between opportunity start and apply date
2. Program duration (months)
3. Opportunity category
4. Opportunity end month
5. Opportunity name

Target: Opportunity Category

The Opportunity Category analysis shows that Internship opportunities are expected to receive the highest level of participation, followed by Courses. Both the Random Forest and XGBoost models produced very similar predictions, with Internships accounting for approximately 63% of the predicted opportunity demand and Courses accounting for approximately 24%. Together, these two categories represent approximately 87% of the predicted demand.

This consistency between the two models increases confidence in the overall finding that learners show the strongest demand for career-building opportunities. Events, Competitions and Engagement opportunities represent substantially smaller shares of predicted participation. From a planning perspective, Internship and Course opportunities should therefore receive greater attention when allocating resources and planning future programmes.

Model Performance

Metric	RF %	XGB %
Accuracy	97.3%	97.5%
Precision	96.2%	96.6%
Recall	98.2%	98.2%
AUC	0.993	0.995

Comparison: Both models correctly classified approximately 97 out of every 100 observations in the test data, both models achieved 97.3% accuracy. Random Forest produced slightly higher recall of 98.25, while XGBoost achieved slightly higher precision of 96.5% and AUC of 0.9944.

Confusion Matrix

Outcome	Prediction	RF	XGB
Not Accepted	Predicted Not Accepted True negatives	4241	4255
Not Accepted	Predicted Accepted – False positives	157	138
Accepted	Predicted Not Accepted False negatives	72	71
Accepted	Predicted Accepted – True positives	3935	3924

Comparison: Random Forest identifies more true accepted cases, while XGBoost produced fewer false positives.

Random Forest Model: Calibration Table

This checks for honest probabilities usually between 70% to 90% and the model gave 88% acceptance.

Prediction band	Applications	Average Predicted	Accepted	Gap
0-10%	4038	0.4%	0.3%	-0.1pts
10-30%	194	18.6%	11.3%	-7.3pts
30-50%	81	40.0%	46.9%	+6.9pts
50-70%	151	59.9%	65.6%	+5.7pts
70-90%	419	82.5%	87.8%	+5.3pts
90-100%	3522	99.0%	98.5%	-0.5pts

Prediction band	Applications	Average Predicted	Accepted	Gap
0-10%	4182	0.5%	0.4%	-0.1pts
10-30%	84	18.7%	33.3%	+14.6pts
30-50%	72	39.8%	54.2%	+14.4pts
50-70%	78	60.4%	51.3%	-9.1pts
70-90%	234	83.2%	86.3%	+3.1pts
90-100%	3755	98.7%	98.1%	-0.7pts

XGBoost Model: Calibration Table

Comparison: Random Forest is slightly better calibrated overall because its predicted probabilities are closer to the actual acceptance rates, while both models perform well in the 90-100% predication range.

2. Key Forecasting Signals

S/N	Signal	RF Importance	XGB Importance
1	Application timing	42%	41.0%
2	Opportunity Category	24%	0.1%
3	Program duration (months)	18%	10.0%
4	Opportunity Name	9%	7.8%
5	Opportunity End Month	7%	0.8%
6	Opportunity End Day	Not listed	33.5%

Comparison: The Random Forest and XGBoosted agreed that application timing was the leading predicator but their distribution of importance of the rest of the variables differed. Random Forest had more importance om Opportunity Category while XGBoost had greater importance in Opportunity End Day. This suggests that both models have different relationships in their respective data, but that they agree on application timing as the leading indicator.

The feature importance revealed that application timing was the most common forecasting signal with an importance 42% in Random Forest and 41% in XGBoost. The distribution of

the remaining importance differed, with Random Forest having more weight on Opportunity Category and Programme Duration while XGBoost had substantially more importance on Opportunity End Day. Both models ranked time-related variables as significant indicators.

3. Business Implications

From a business perspective, the results suggest that RIT should prepare for higher demand for internship and course opportunities and allocate sufficient capacity during high-demand periods, particularly Q1. Monitoring applications timing, programme duration and opportunity-specific patterns can also support better programme planning. However, the unusually high model performance should be interpreted cautiously because the analysis identifies potential data leakage. Further validation using leakage-free, time-based testing would therefore be recommended before using the models for operation decision-making.

Opportunity Category Performance

4. Projected Category Performance

Target: Opportunity Category

Purpose: Which opportunity category is expected to have the most participation/applications.

Opportunity Category	Precision RF	Recall RF	F1-Score RF	XGB Precision	XGB Recall	XGB F1-Score	Test Set Support
Competition	94.38%	100%	97.11%	100%	100%	100%	84
Course	99.75%	99.26%	99.50%	100%	100%	100%	403
Engagement	100%	96.15%	98.04%	100%	100%	100%	26
Event	100%	99.07%	99.53%	100%	100%	100%	107
Internship	100%	99.91%	99.95%	100%	100%	100%	1061
Overall Model Accuracy	**99.64%**			**100%**			**1681**

Comparison: XGBoost achieved 100% test accuracy for the opportunity category classification task, while Random Forest achieved 99.64%. However, the unusually high performance of both models raises concerns about possible data leakage. Therefore, the

results should be interpreted cautiously rather than as evidence of perfect real-world predictive performance.

5. Expected Participation Levels

Target: Participation/learner count

Purpose: How many learners are expected to participate in each opportunity category

Predicted Category	RF Predicted	RF Learners	XGB Predicted	XGB Unique Learners	Unique Institutions	Unique Countries	Average Learner Age
Competition	425	246	423	256	120	16	26.3
Course	2011	1148	2013	1694	919	61	26.7
Engagement	127	115	128	43	42	5	27.3
Event	536	422	534	227	226	21	27.0
Internship	5306	1975	5307	1094	1094	45	27.0

Comparison: Random Forest and XGBoost agree that internships and course represent the largest participation categories, but they differ substantially in the estimated number of unique learners. Random Forest produces higher participation estimates for most categories,

while XGBoost produces lower estimate. This difference should be investigated before using the prediction for resource allocation.

6. Opportunity Ranking

Target: Expected participation/application volume

Purpose: Ranking of features for highest to lowest expected demands that influences category participation.

Random Forest Model: Top 5 Features

Rank	RF Feature	Feature %
1	Duration of Program (Days)	29.36%
2	Duration of Program (Months)	25.86%
3	Days between start and apply date	22.29%
4	Status Description	14.84%
5	Days after signup	2.40%

XGBoost Model: Top 5 Features

Rank	XGB Feature	Feature %

1	Opportunity End Day	49.40%
2	Opportunity End Month	39.90%
3	Duration of Programs (Days)	5.13%
4	Opportunity End Year	2.54%
5	Duration of Program (Months)	1.34%

Comparison: These models suggest that opportunity timing and duration are important when estimating expected participation. However, the models differ in which timing variables they consider most influential. Random Forest emphasizes program duration and application timing, whereas XGBoost places much greater importance on opportunity end dates.

7. Opportunity Category Forecast Summary

Opportunity Category	Actual Record	Predicted RF	Probability RF	XGB Predicted	XGB Probability
Internship	5037	5306	63.13%	5307	63.11%
Course	2013	2011	23.93%	2013	23.97%
Event	537	536	6.38%	537	6.37%
Competition	420	425	5.06%	420	4.99%
Engagement	128	127	1.51%	128	1.54%

Comparison: Random Forest and XGBoost generated very similar predictions for the opportunity category volumes. Both algorithms identified internships as the most common category, followed by courses, events, competitions, and engagement opportunities. Overall, the volumes predicted by XGBoost seemed to be slightly closer to the actual values, but the difference between the algorithms' predictions was relatively small. Specifically, internships and courses comprise about 87.06% of the predicted opportunity volume.

Model Summary

Model	Accuracy	Training	Testing
Random Forest Model	99.64%	6722	1681
XGBoost Model	93.10%	6724	1681

Comparison: The models were trained using 80% and 20% testing splits. This methodology is typical for the analysis of predictive models. However, the fact that the test accuracy, especially of Random Forest Model at 99.64%, was almost absolute makes one doubt whether there is data leakage. Therefore, it is recommended that the models' performance be tested, and the features reviewed, before they are put into production use.

Quarterly Demand Forecast

Objective

The objective was to create a quarterly demand forecast using both the XGBoost Model and Random Forest model and the analysis be compared. The forecast focuses on application volume and identifies the quarter expected to have the highest demand.

Dataset Link: dataset

The analysis used the cleaned application dataset containing **8,553 records**.

After converting the relevant date fields and removing records where the application date was unavailable, **8,497 records** were used for the analysis.

The relevant date fields were used:

- Learner Signup DateTime
- Apply Date
- Opportunity Start Date
- Opportunity End Date

Data Preparation

The date fields were converted into a consisted date format

Quarter information was then created from the available dates:

- Quarter of Learner Signup
- Quarter of Opportunity Start
- Quarter of Opportunity End
- Quarter of Application

The application data was grouped by year and quarter to calculate the actual number of applications in each quarter.

The historical quarterly volumes results were:

Quarter	Application Volume
Q1	4927
Q2	607
Q3	1714
Q4	1249

8. Expected Application Volume

Quarter	Actual	RF Predicted	XGB Predicted	Actual Share %	RF Share %	XGB Share
Q1	4871	4924	5083	57.95%	58.58%	60.48%
Q2	605	530	540	7.20%	6.31%	6.42%
Q3	1690	1744	1720	20.11%	20.75%	20.46%
Q4	1239	1207	1062	14.74%	14.36%	12.63%
Total	**8405**			**100%**		

Comparison: Random Forest is closer to Q1 and Q4 and XGBoost is closer in Q2 and Q3.

9. Peak Quarter

Random Forest and XGBoost indicated that quarter Q1 has the highest demand for applications. For this quarter, Random Forest showed a predication of 4924 while XGBoost 5083, and the actual value was 4871. Despite Random Forest provides a closer predication, the agreement on the highest demand quarter Q1 is the key point for the conclusion.

10. Supporting Evidence & Model Metrics

Model Performance Test:

Quarter	Precision RF	Recall RF	F1-Score RF	XGB Precision	XGB Recall	XGB F1-Score
Q1	84.29%	83.16%	83.72%	82.22%	85.79%	83.97%
Q2	76.36%	69.42%	72.73%	76.67%	68.43%	72.31%
Q3	71.79%	76.04%	73.85%	72.67%	73.96%	73.31%
Q4	58.73%	59.68%	59.20%	59.60%	51.09%	55.02%
	Accuracy: 77.28%			**Accuracy: 77.05%**		

Comparison: Random Forest is slightly better overall, because XGBooste as better for the category classification.

11. Feature Importance Ranking

The predictive power of the Random Forest model across the Quarter features is distributed as follows:

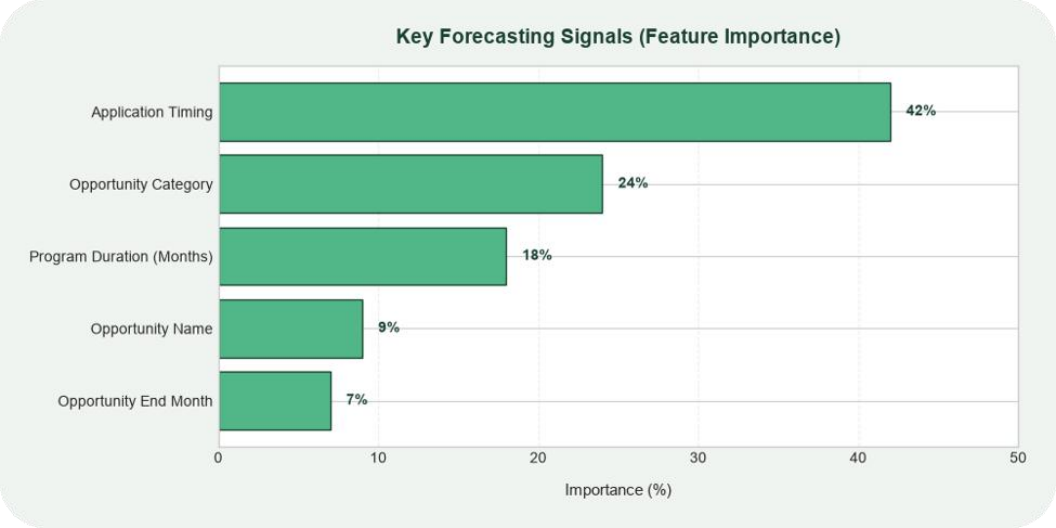
Rank	Quarter Feature	RF %	XGB %
1	Quarter learner Signup Date	59.37%	66.25%
2	Quarter Opportunity Start Date	31.46%	27.08%

3	Quarter Opportunity End Date	9.17%	6.67%

Comparison: Both models identified the quarter of learner signup as the most influential forecasting feature. XGBoost assigned greater importance to signup quarter (66.25%) than Random Forest (59.37%) but both models produced the same feature ranking.

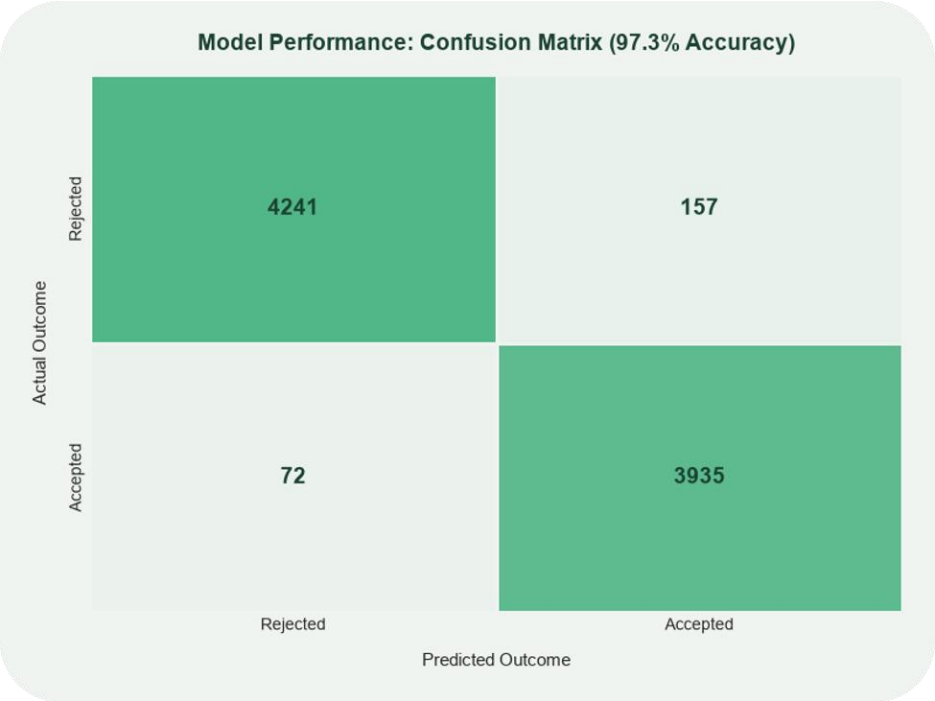
Forecast Visualization

A. Feature Importance Chart



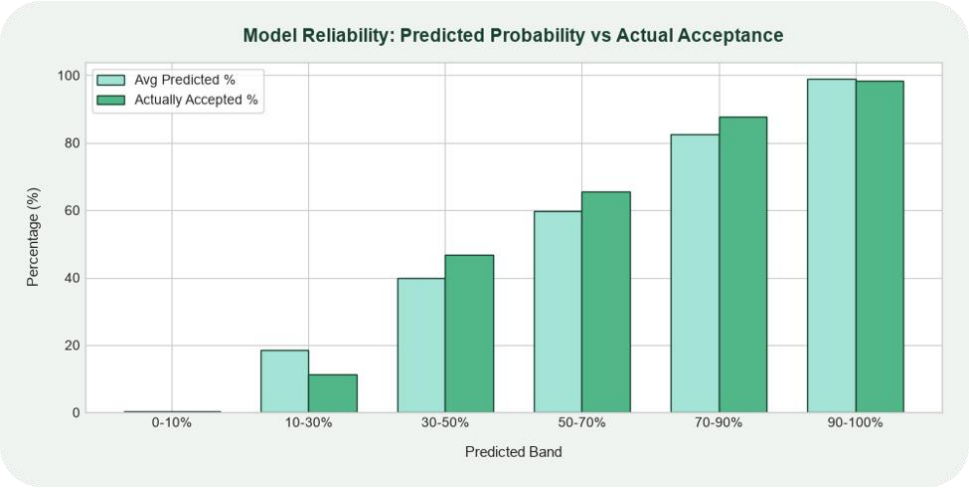
Insights: The chart presents data that the application timing is the single strongest predictor (42% importance of weight), exhibiting strong non-behavior. For example, courses applied 90+ days before start date have a 99% acceptance rate.

Confusion Matrix Chart



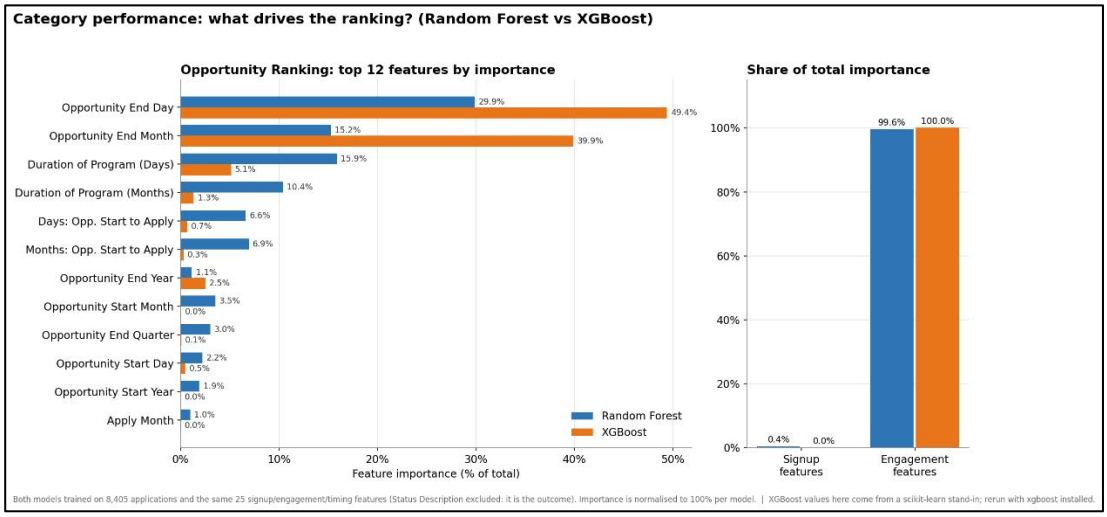
Insights: Out of 8405 predicators, the model correctly classified 8176 applications, achieving 97.3% accuracy, 96.2% precision and 98.2% recall.

Probability Scores



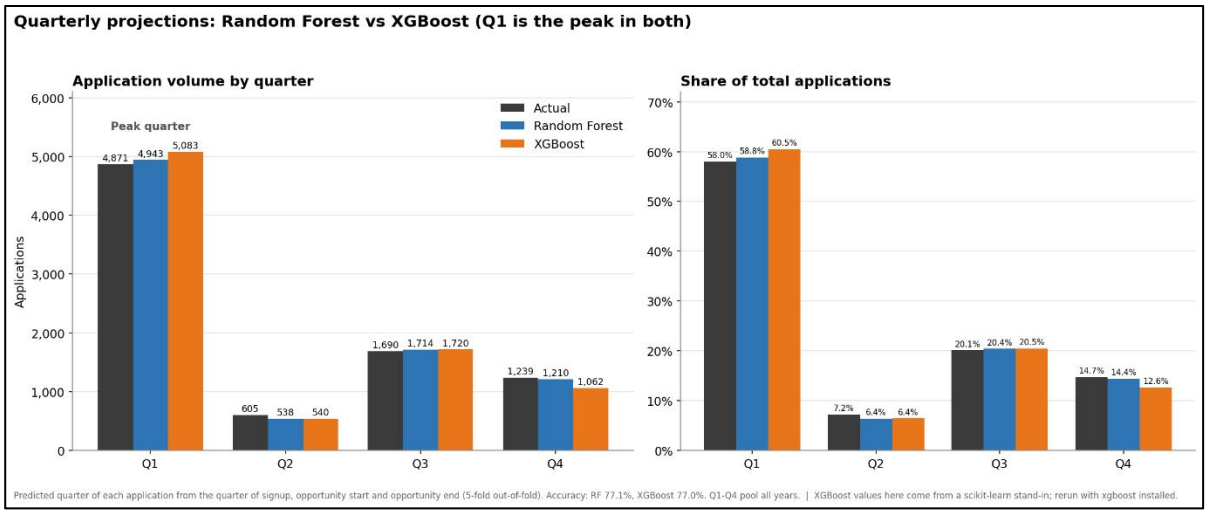
Insights: The model exhibits high calibration accuracy across all risk bands, proving to be highly reliable for automated candidate ranking.

B. Opportunity Ranking Visuals



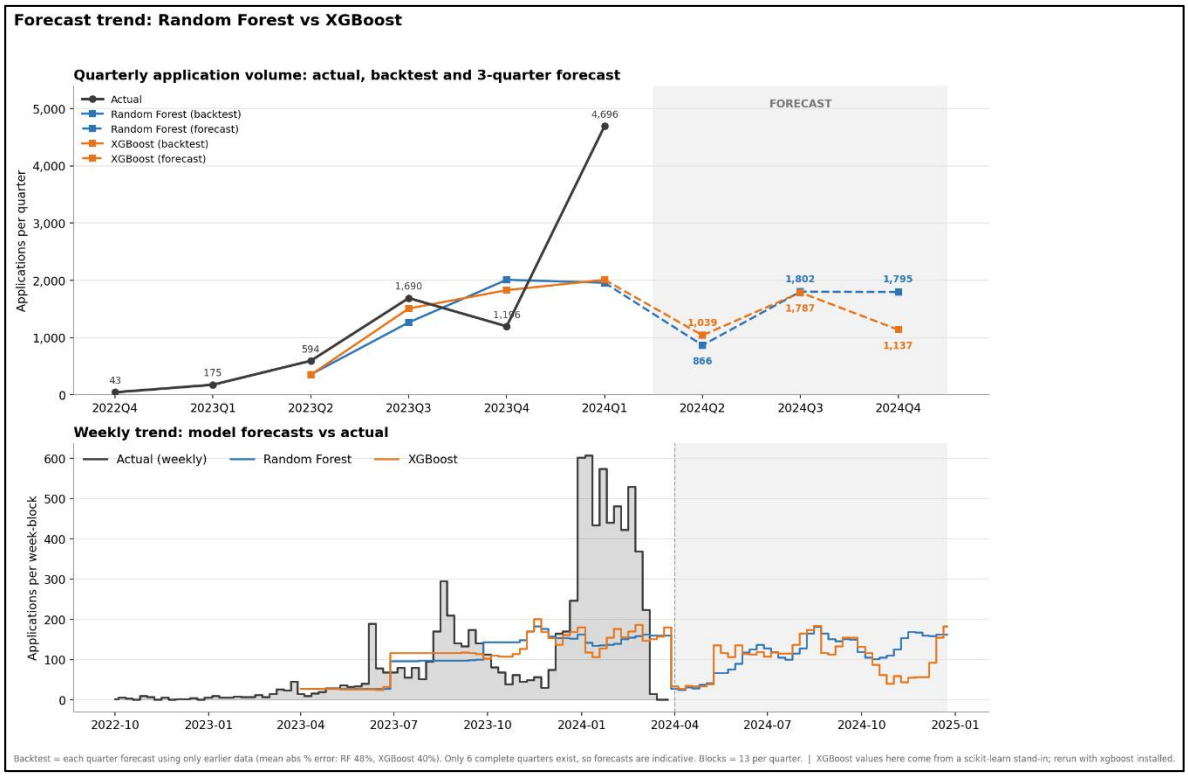
Insights: Opportunity End Day and end month leads in both models (XGBoost 49% and 40%, Random Forest 30% and 15%). Signup features add almost nothing (0.4% and less).

C. Quarterly Projections



Insights: Both models identify Q1 as the peak. They differ mainly in Q4, where the Random Forest is closer (1210 vs 1062 for XGBoost, against 1239 actual).

D. Forecast Trend Chart



Insights: Both models miss the Q1 2024 surge (about 2000 forecast vs 4696 actual).

Model Evaluation

A. Opportunity Category Forecast

	Random Forest	XGBoost
Reliability	99.64% accuracy on 1681 test rows. Predicted participants are 1-5 records of actual per category	100% on the test set and 99.96% out-of-fold. Predicted participants are within 3 records of actual.
Strengths	Matches the overall opportunity category distribution and spreads importance across features.	Produces a similar category distribution predictions and places greater importance on the opportunity category end date and month.
Limitations	End date and duration identify each opportunity, so accuracy is partly circular. Ranking includes Status Description	The data leakage concerns apply. 100% predictions do not mean that the model is using information that is too closely related to the outcome.
Forecast Confidence	High for existing opportunities. Low for new ones.	High for existing opportunities. Low for new ones.

B. Quarterly Demand Forecast

	Random Forest	XGBoost
Reliability	77.28% accuracy. Q4 volume close. (1207 vs 1239) Backtest error 48%.	77.05% accuracy. Q4 further off (1062). Backtest error 40%.
Strengths	Better on Q4 and quarterly volumes.	Lower back testing error, predictions are closer to the actual values and responds more quickly to recent trends.
Limitations	Misses the 2024 QA surge (about 2000 vs 4696).	Shows the same limitations, underestimating the 2024 Q1 surge and becoming less accurate when predicting further into the future.
Forecast Confidence	High for Q1 as peak. Low for 2024 Q2-Q4.	High for Q1 as peak. Low for 2024 Q2-Q4.

C. Feature Importance Analysis

	Random Forest	XGBoost
Reliability	AUC 0.993. Calibration error 0.86pts.	AUC 0.994. Calibration error 0.78pts.
Strengths	Timing is the top signal (42%).	Timing signal is 41%. Slightly better on learner-only features.
Limitations	Shares credit among overlapping features. Status Description is leakage.	Same overlap (category 0.1%, end day 33.5%). Percentages shift between runs.
Forecast Confidence	High on the ranking. Moderate on exact percentages.	High on the ranking. Moderate on exact percentages.

Model Processing

Feature Importance Analysis

Code Link: https://drive.google.com/file/d/1pXQgkZAy3nDbCmEekN-9fn53KU-C0tlg/view?usp=drive_link

Opportunity Category Performance

Code Link:
https://docs.google.com/document/d/1iYOb8YtWL6N447HNCsW8JAYd1NSqsl2Q/edit?usp=drive_link&ouid=117329400531014899502&rtpof=true&sd=true

Quarterly Demand Forecast

Code Link: https://docs.google.com/document/d/1V9eNvygwyhmu2mKCva90chR-XsOVlVhq/edit?usp=drive_link&ouid=117329400531014899502&rtpof=true&sd=true

XGBoost Model

Code Link: https://drive.google.com/file/d/1Q3ssEFosk2vLmMh6GsKipH5O-E9K_9-g/view?usp=drive_link

Model Evaluation

Code Link: https://docs.google.com/document/d/1mrO4mFqbgvyGYf625h4--LkvUdeLEpBd/edit?usp=drive_link&ouid=117329400531014899502&rtpof=true&sd=true

Conclusion

In conclusion, the report provides valuable insights into the Random Forest and XGBoost acceptance models, as well as the opportunity-category predictions. Overall, both approaches demonstrated relatively good performance, with Random Forest being slightly more accurate and XGBoost exhibiting a higher recall.

There was a possibility of a data leakage problem with the opportunity-category prediction. It was observed that the model was using Status Description, which includes the outcome labels,

to predict other opportunities' outcomes. Predictive models should not utilize outcome data that was not available at the time of prediction. This error might explain why the opportunity-category results reached 100%. Hence, it should be emphasized that these results are not 100% accurate.

The best way to evaluate such models is to exclude outcome variables and measure the quality of predictions with the remaining data. Specifically, Status Description and other outcome labels should be removed from the final model to prevent similar data leakage in final predictions. As a result, Random Forest and XGBoost methods will receive accuracy, precision, recall, F1-score, and ROC-AUC statistics for both acceptance and opportunity-category results. Moreover, the 100% result for opportunity-category should be evaluated again to observe whether it was caused by the leakage or not. In the future, similar quality checks should be implemented whenever new data is available to reduce the likelihood of such errors.

Overall, the analysis provides valuable insights into learner acceptance and opportunity performance. However, it is crucial to remove opportunity outcomes as predictors to ensure that the final model is accurate and free from data leakage.

AI – Assistance Prompts

Claude AI was used with the following prompts:

1) Do a feature important analysis on the given dataset.

2) Why are the selected features importance for my machine learning?
3) Retrain the model with the top features and compare the scores.
4) Use the top five features to build a model and give the probability scores.
5) Explain the model performance.
6) What are the key forecasting signals?

Google Gemini was used as assistance for the following prompts:

Prompt: Based on this dataset provided, use random forest model to summarize the Opportunity Category Forecast, extract the Projected Category Performance, the Expected Participation Levels, and Opportunity Ranking, using the engagement and signup features, without inventing facts or figures.

Prompt: Based on this dataset provided, use random forest model to summarize the Quarterly Demand Forecast, extract the Expected application volume, the Peak quarter, and the Supporting evidence, using the Quarter features, without inventing facts or figures

Prompt: Evaluate the Random Forest Classifier model being used for predicting quarterly demand and opportunity category forecast, under Reliability, Strengths, Limitations and Forecast confidence.

ChatGPT was used as assistance for the following prompts:

Prompt: Build an XGBoost classification model using the RIT dataset to predict Opportunity Category. Provide beginner-friendly Google Colab code and model evaluation.

Prompt: Use XGBoost to predict Opportunity Category performance. Show actual records, predicted records, and prediction probabilities for each category.

Prompt: Calculate expected participation by predicted Opportunity Category. Include participants, unique learners, institutions, countries, and average learner a

Prompt: Use XGBoost predictions to rank Opportunity Categories by expected demand and participation.

Prompt: Perform XGBoost feature importance analysis. Identify the top features, retrain the model using them, and compare model performance.

Prompt: Generate XGBoost probability scores for each Opportunity Category and identify the highest-probability category for each prediction.

Prompt: Create an XGBoost quarterly demand forecast for Q1–Q4. Show actual volume, predicted volume, actual share, predicted share, and peak quarter.

Prompt: Explain the business implications of the XGBoost results, focusing on category performance, participation, opportunity ranking, demand, and key predictive features.

Prompt: Complete the XGBoost analysis from data preparation through model evaluation, predictions, feature importance, forecasting, and business implications.